PAPER_342 ■ Magnetar 7-Component DPM-THz Frequency Form: S26 Spin-Down Plus THz Modes

Author: Daniel T. Murphy

Session: *96*

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Abstract

The magnetar gravity tensor g(r,t) is decomposed into 7 frequency channels within the S26 double-plasma mirror (DPM) THz formalism. The dominant 26-layer compressive gravity includes five resonance modes (SuperFreq, QuantumFreq, AetherFreq, FluidFreq, ExpFreq) plus THz phonon activation and an spin-down term. For SGR J1745-2900 class magnetars with P = 3.76 s, B = 2■10■■ T, the magnetic energy reservoir is M_mag = 2.01■10■7 J.

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2. Core Physics

2.1 7-Component S26 Gravity Form

```
g(r,t) = \sum_{i=1}^{26} \left[ a_i^{\rm DPM} + a_i^{\rm THz} + a_i^{\rm SF} + a_i^{\rm QF} + a_i^{\rm AF} + a_i^{\rm FF} + a_i^{\rm EF} \right]
```

Components:

- **DPM baseline**: `a_i^{\rm DPM} = G M_i / r_i^2 \cdot [UA]_i`
- **THz phonon**: `a_i^{\rm THz} = a_0^{\rm THz} \cdot \cos(2\pi f_{\rm THz} t) \cdot Q_i`
- **SuperFreq** (SGR 1745): `a_i^{\rm SF} = \alpha_{\rm SF} \cdot \omega_{\rm SF}^2 \cdot r \cdot [SCm]_i`
- **QuantumFreq**: `a_i^{\rm QF} = \hbar \omega_{QF} / (m r^2) \cdot f_i`
- **AetherFreq**: `a_i^{\rm AF} = \rho_{\rm UA} \cdot c_s^2 / r \cdot \sin(\omega_{\rm AF} t)`
- **FluidFreq** (Navier-Stokes): `a_i^{\rm FF} = \eta \nabla^2 v / \rho`
- **ExpFreq** (Hubble expansion): `a_i^{\rm EF} = H_0 \cdot \dot{r}_i`

2.2 Spin-Down Rate

```
\dot{\nu} = -\frac{f_{\rm react}}{2\pi P}
```

where P = 3.76 s (period), and f_react is the UQFF vacuum reactance frequency.

2.3 Magnetic Energy Reservoir

```
M_{\rm mag} = \frac{B^2 V}{2\mu_0} = \frac{(2\times 10^{10})^2 \cdot V_{\rm mag}}{2\mu_0}
= 2.01\times 10^{37}\ \mathrm{J}
```

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3. Key Values

Quantity	Formula	Value
Period	P	3.76 s
Surface field	B	2■10■■ T
Magnetic energy	B■V/(2■0)	2.01■10■7 J
Spin-down rate	?? = -f_react/(2pP)	-6.7■10?■■ Hz/s
THz frequency	f_THz	~1 THz
26-layer sum	S26 contributions	7 channels per layer

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4. Physical Significance

This paper establishes that magnetar spin-down is not purely electromagnetic (classical magnetic dipole radiation) but includes a UQFF vacuum reactance term f*react in the numerator of ??. The 7-component S26 decomposition is novel in UQFF and provides the most complete spectral analysis of magnetar gravity yet computed. The THz mode connects magnetar physics to laboratory cold-physics (H2O/H2 quantum transitions identified in PAPER*339).

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5. Deduplication Note

- **vs. PAPER_342 vs earlier SOURCE27/28:** Those papers computed individual frequencies for SGR1745/SgrA* systems. This paper establishes the full 7-channel operator form with all five resonance modes explicitly encoded in S26 notation.
- **vs. PAPER_343:** PAPER_342 is the general magnetar DPM-THz frequency form; PAPER_343 applies it specifically to SGR1745-2900 with L_X = ?_vac■f_res■V.

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6. Classification

Physics Territory: FIRST magnetar S26 7-component DPM-THz frequency decomposition **Scale:** Stellar (neutron star magnetar) **CP Implementation:** `MagnetarDPMTHzFrequencyFormCalculator` (CondensedPhysics3.py, Session 96)

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]■exp(-?■?t) = 1 - 5.7e-1 ■ exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s■.
